

Diagonal Pair Coulomb Density Matrix on a Sphere

Riccardo Fantoni*

Università di Trieste, Dipartimento di Fisica, strada Costiera 11, 34151 Grignano (Trieste), Italy

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We tabulate, on the diagonal, the n th power of the primitive approximation for the pair Coulomb density matrix on a sphere and we discuss its behavior at particles contact as n grows. Our results show that even on a curved surface quantum statistical physics is fundamentally different from its classical counterpart. If in the latter one needs to artificially introduce a regularization of the divergent attractive Coulomb potential preventing particles contact, for example with a hard core, in the former this is not necessary.

Keywords: Pair Coulomb Density Matrix; Primitive Approximation; End-Point Approximation; Sphere; Cusp Condition

CONTENTS

I. Introduction	1
II. Approach to the continuum for the pair Coulomb density matrix	2
III. Cusp condition	5
IV. Conclusions	6
A. $\rho_{i0}(\mathbf{0}, \mathbf{0}; t)$	6
B. The code	6
Author declarations	8
Conflicts of interest	8
Data availability	8
Funding	8
References	8

I. INTRODUCTION

In a quantum many body (Monte Carlo) simulation of a Coulomb liquid it is often useful to have access to the low temperature pair Coulomb density matrix ρ . In the high temperature classical limit the *primitive approximation* $\rho_{\text{primitive}}$ becomes exact. Within Path Integral Monte Carlo (PIMC) of a many body system [1] in order to access the quantum regime with less imaginary time discretization error it is important to use the *pair-product inter-action*. The imaginary time t is related to the temperature T of the liquid by $t = \beta = 1/k_B T$ with k_B Boltzmann constant. The main ingredient to construct the pair-product inter-action is the inter-action of the pair density matrix or the *pair-inter-action*. Starting from $\rho_{\text{primitive}}$ at a high temperature T_0 we can halve the temperature simply by squaring it ¹ [2, 3] (see also section IV.G of Ref. [1]). This lower temperature pair density matrix already possesses different properties than the primitive approximation. In position representation, on the diagonal

$$\left| \frac{\phi(\mathbf{r}_1, \mathbf{r}_2)}{k_B T_0} \right| = \left| \ln \left[\frac{\rho_{\text{primitive}}(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)}{\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)} \right] \right| \xrightarrow{\mathbf{r}_1 \rightarrow \mathbf{r}_2} \infty, \quad (1.1)$$

* riccardo.fantoni@scuola.istruzione.it

¹ The squaring method can be used if an exact solution is known, so that successive squarings allow to reach temperatures $T_0/2^n$ for $n = 1, 2, 3, \dots$

where $\phi(\mathbf{r}_1, \mathbf{r}_2)$ is the pair Coulomb potential between particle 1 at $\mathbf{r}_1$ and particle 2 at $\mathbf{r}_2$. On the other hand, for a repulsive Coulomb potential, when Trotter formula holds [?], we can reach the temperature $T_0 = T_n/n$ sending $n \rightarrow \infty$ and $T_n \rightarrow \infty$, i.e. through the *continuum limit*². In this case we find

$$|P(\mathbf{r}_1, \mathbf{r}_2; 1/k_B T_0)| = \lim_{n \rightarrow \infty} \left| \ln \left[\frac{[\rho_{\text{primitive}}]^n(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; n/k_B T_n)}{[\rho_0]^n(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; n/k_B T_n)} \right] \right| \xrightarrow{\mathbf{r}_1 \rightarrow \mathbf{r}_2} \text{finite}, \quad (1.2)$$

where we denote with $[\rho]^n$ the n th power of the pair density matrix: $n = 2$ for the *square*, $n = 3$ for the *cube*, and so on. Moreover a *cuspl condition* must hold for P as the two particles approach each other, like in the flat space case [1, 4, 5].

The same contact value smearing property holds at any temperature T_0 . This property makes it of fundamental importance not to use the primitive approximation for example in a simulation of an electron proton binary mixture where the divergence of the attractive Coulomb potential between unlike species at contact would inevitably let the fast electrons stick onto the slow protons, and consequently spoil the simulation. In other words quantum statistical physics does not need hard core regularization which is otherwise an essential but artificial feature in classical statistical physics necessary to avoid unlike particles collapse.

In this work we will show how this purely quantum smearing effect can be accomplished on a Sphere by approaching the $n \rightarrow \infty$ continuum limit. We will not give the exact solution for the pair Coulomb density matrix as was done in flat space in Refs. [5–7] but we will give an approximate numerical solution instead starting from the classical high temperature limit primitive approximation. If we were able to find an exact solution of the pair Coulomb density matrix on a sphere we would not need to worry about deriving it from the primitive approximation. In absence of an analytic solution here we propose to use the path integral discretized in imaginary time with timestep $\tau = 1/k_B T_n$ to reach a temperature $T_0 = 1/nk_B \tau$ approaching the continuum limit $\tau \rightarrow 0$. Of course this limit cannot be reached numerically but we will be able to say something about the trend as n increases.

We will only worry about the *end-point approximation* (see section IV.F of Ref. [1]) to the pair Coulomb density matrix, i.e. the diagonal component of the pair-inter-action. Even if this is exact on a flat three dimensional space due to the peculiar properties of the Coulomb pair potential [5] it becomes just an approximation on the Sphere.

II. APPROACH TO THE CONTINUUM FOR THE PAIR COULOMB DENSITY MATRIX

We show a method to lower the temperature of the hot Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work in flat space of Refs. [5–7] to a curved surface.

We will denote with a Greek index a coordinate component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius a is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \cos^2 \theta \end{pmatrix}. \quad (2.1)$$

with $g(\mathbf{r}) = \det||g_{\alpha\beta}|| = a^4 \cos^2 \theta$ and $\theta \in [-\pi/2, \pi/2]$ the polar angle measured from the equatorial plane, so that $\theta = \pi/2$ is the north pole and $\theta = -\pi/2$ is the south pole.

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ of the density matrix $\rho = \exp(-Ht)$, where H is the Hamiltonian operator for the two bodies and t is the imaginary time that in statistical physics is identified with the inverse temperature $\beta = 1/k_B T$ with k_B Boltzmann constant. These are initially at $\mathbf{r}_1$ and $\mathbf{r}_2$ and at an imaginary time t later they are at $\mathbf{r}'_1$ and $\mathbf{r}'_2$.

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = \left(\frac{\lambda_1}{\sqrt{g(\mathbf{r}_1)}} \nabla_{1\alpha} \sqrt{g(\mathbf{r}_1)} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g(\mathbf{r}_2)}} \nabla_{2\alpha} \sqrt{g(\mathbf{r}_2)} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (2.2)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g(\mathbf{r}_1) g(\mathbf{r}_2)}, \quad (2.3)$$

² For any finite n one can treat the attractive Coulomb as well as the repulsive Coulomb.

where, for particle $i = 1, 2$, $\lambda_i = \hbar^2/2m_i$ with m_i the mass of particle i , $g(\mathbf{r}_i) = a^4 \sin^2 \theta_i$, $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial\theta_i, \partial/\partial\varphi_i)$ is the gradient respect to the coordinates $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$, δ is a two dimensional Dirac delta function, ρ is the pair density matrix, and ϕ is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (2.4)$$

where $Z_i e$ is the charge of particle i and for the distance between the two points $\mathbf{r}_1$ and $\mathbf{r}_2$, $d(\mathbf{r}_1, \mathbf{r}_2)$, on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = a \arccos[\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (2.5)$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2a \sin(d_g/2a). \quad (2.6)$$

For particles living **in** the surface [8] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\mathbf{r}_1, \mathbf{r}_2)/\ell], \quad (2.7)$$

with ℓ a constant with the dimensions of a distance. We will call ϕ_{on}^g , ϕ_{on}^e , ϕ_{in} the three cases just illustrated.

We will measure energy in units of the Hartree e^2/a_B , with e the charge of the electron and $a_B = \hbar^2/m_e e^2$ Bohr radius, with m_e the mass of the electron. And length in units of the Bohr radius a_B . We will call $\mu_i = m_i/m_e$ the reduced mass of particle i . Temperatures are in units of $\hbar^2/m_e a_B^2 k_B$.

The free particles, $\phi \rightarrow 0$, pair density matrix can be determined from the work of Bastianelli and Corradini [9]. We will call their solution for the single free particle i , $\rho_{i0}(\mathbf{r}_i|\mathbf{r}'_i; t)$. Therefore the pair free density matrix will be

$$\rho_0(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_{10}(\mathbf{r}_1|\mathbf{r}'_1; t) \rho_{20}(\mathbf{r}_2|\mathbf{r}'_2; t). \quad (2.8)$$

Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (2.9)$$

where P is the *pair-inter-action* [1].

Aim of this work is to tabulate $P(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; t)$ for the three Coulomb cases illustrated above, P_{on}^g , P_{on}^e , P_{in} . In the hot case, for small $t \ll 1$, for each of these cases, the *primitive approximation* [1]

$$P_{\text{primitive}} \approx t\phi, \quad (2.10)$$

holds.

Therefore, introducing a timestep τ , we can find the PDCM at all subsequent timesteps through the following convolution path integral

$$\begin{aligned} \rho(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; n\tau) &= \int \rho(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}_{1,1}, \mathbf{r}_{2,1}; \tau) \rho(\mathbf{r}_{1,1}, \mathbf{r}_{2,1}|\mathbf{r}_{1,2}, \mathbf{r}_{2,2}; \tau) \dots \rho(\mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1}|\mathbf{r}'_1, \mathbf{r}'_2; \tau) \times \\ &\quad \prod_{k=1}^{n-1} \sqrt{g(\mathbf{r}_{1,k})g(\mathbf{r}_{2,k})} d\mathbf{r}_{1,k} d\mathbf{r}_{2,k}, \end{aligned} \quad (2.11)$$

where we denote with $\mathbf{r}_{i,k}$ the position of particle i at timeslice k , $d\mathbf{r} = d\theta d\varphi$, and we choose τ small enough so that

$$\rho(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; \tau) \approx \rho_0(\mathbf{r}_1, \mathbf{r}_2|\mathbf{r}'_1, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2} \quad (2.12)$$

$$= \rho_{10}(\mathbf{r}_1|\mathbf{r}'_1; \tau) \rho_{20}(\mathbf{r}_2|\mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2}, \quad (2.13)$$

is a good approximation. We will perform the $4(n-1)$ integrations in Eq. (2.11) through a simple Monte Carlo (MC) quadrature [10] where a MC point is the tuple $(\mathbf{r}_{1,1}, \mathbf{r}_{2,1}, \dots, \mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1})$. It is important to stress that this is only possible if $Z_1 Z_2 > 0$ otherwise the Coulomb potential will be unbounded from below and the path integral of Eq. (2.11) breaks down [11].

According to the analysis of Ref. [9] we can further approximate

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i|\mathbf{r}'_i; \tau) g^{1/4}(\mathbf{r}'_i) \approx \frac{\mu_i}{2\pi\tau} \exp \left\{ -\frac{\mu_i}{2\tau} \delta_{\alpha\beta} (\mathbf{r}_i - \mathbf{r}'_i)^\alpha (\mathbf{r}_i - \mathbf{r}'_i)^\beta - \tau \frac{V_R(x_i) + V_R(x'_i)}{2} \right\}, \quad (2.14)$$

where $\delta_{\alpha\beta}$ is the Kronecker delta, the normalization $2\pi\tau/\mu_i$ is just the one of a free particle of mass μ_i on a plane and corresponds to the exact path integral performed with an action $\int_0^\tau dt' \mu_i \delta_{\alpha\beta} \dot{\mathbf{r}}_i^\alpha \dot{\mathbf{r}}_i^\beta / 2$, $x_i^2 = \delta_{\alpha\beta} \mathbf{r}_i^\alpha \mathbf{r}_i^\beta$, and V_R is the effective potential due to the curvature of the Sphere, $R = 2M^2$ with $M = 1/a$, namely

$$V_R(x) = -\frac{M^2}{6} - \frac{5(Mx)^2 - 3 + [(Mx)^2 + 3] \cos(2Mx)}{48x^2 \sin^2(Mx)}, \quad (2.15)$$

which is equation (2.32) of Ref. [9].

We may additionally introduce the following exact Feynman-Kac expression for the one free body density matrix at time t

$$g^{1/4}(\mathbf{r}_i) \rho_{i0}(\mathbf{r}_i | \mathbf{r}'_i; t) g^{1/4}(\mathbf{r}'_i) = \langle e^{-S_R} \rangle, \quad i = 1, 2 \quad (2.16)$$

$$S_R = \int_0^t dt' V_R(x_i) \quad (2.17)$$

where $\langle \dots \rangle$ denotes an average over all single particle Gaussian random walks from $\mathbf{r}_i$ to $\mathbf{r}'_i$ in a time t . In Appendix A we give the diagonal part $\rho_{i0}(\mathbf{r}_i | \mathbf{r}_i; t) = \rho_{i0}(\mathbf{0} | \mathbf{0}; t)$ up to orders t^8 . Our Eq. (A1) coincides with the perturbative expansion (3.23) of Ref. [9] for $d = 2$ spatial dimensions.

Even if in general we will be interested in the full $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$ in practice in a many body simulation we may use the *end point approximation* for the inter-action as shown in section IV.F of Ref. [1] where one only needs the diagonal components $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = P(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t)$. Where on a sphere the geodesic distance $d_g(\mathbf{r}_1, \mathbf{r}_2)$ between any two points $\mathbf{r}_1$ and $\mathbf{r}_2$ is bounded from above by πa . Due to the Sphere symmetry, without loss of generality, we may choose $\mathbf{r}_1 = (\pi/2, 0)$ and $\mathbf{r}_2 = (\theta, 0)$, so that $d_g = a(\pi/2 - \theta)$.

On the diagonal we will have

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t) = \rho_0(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t) e^{-P(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t)}, \quad (2.18)$$

we will then tabulate

$$P(m\delta; n\tau) = -\ln \left[\frac{\rho(m\delta; n\tau)}{\rho_0(m\delta; n\tau)} \right], \quad n = 1, 2, 3, \dots, \quad m = 1, 2, 3, \dots, m_{\max} \quad (2.19)$$

for given small τ and δ so that $m_{\max} = \pi/\delta$. For an imaginary time $t_n = n\tau$ this will require a number of $4(n-1)$ quadratures for each m to calculate $\rho(m\delta; n\tau) = \rho(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t_n) = \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$ and $\rho_0(m\delta; n\tau) = \rho_0(d_g(\mathbf{r}_1, \mathbf{r}_2)/a; t_n) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$ for $m = 1, 2, 3, \dots, m_{\max}$, through the corresponding path integral convolution (2.11). Each convolution requires 4 quadratures over the azimuthal angle $\varphi \in [-\pi, \pi[$ and the polar angle $\theta \in [-\pi/2, \pi/2]$ of each particle. In any case, in a many body simulation the pair-product inter-action is needed at a fixed small t , which determines the discretization error [1]. So one needs to tabulate the PDCM just once, at the beginning of the simulation.

In Fig. 1 we plot the ratio $R(m\delta; t) = P(m\delta; t)/t\phi(m\delta)$ as a function of m for $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2, \delta = 0.03$ and $n = 2, 3, 4, 5$. We see how at large particles distance the ratio approaches 1 but at contact it tends to $\approx 1/n$. Therefore we expect that at constant $t = n\tau$ as $n \rightarrow \infty$, in the $\tau \rightarrow 0$ continuum limit, the ratio $R(0; t) \rightarrow 0$. Then we can say that $P(d_g \rightarrow 0; t)$ must be an infinite of order less than the infinite of $\phi(d_g \rightarrow 0)$. Moreover the PCDM at particles contact must obey to a cusp condition [5] as discussed in the next section.

In Table I we show $P(0; t)$ and $R(0; t)$ for $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2$ and increasing n . The value at contact, $d_g = 0$, is obtained through a linear extrapolation of the first two points at $m = 1, 2$ of the top panel of Fig. 1.

TABLE I. Table for $P_{\text{on}}^g(0; t)$ and $R_{\text{on}}^g(0; t) = \lim_{\delta \rightarrow 0} P_{\text{on}}^g(\delta; t)/t\phi_{\text{on}}^g(\delta)$ for $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2$. The value at contact, $d_g = 0$, is obtained through a linear extrapolation of the first two points at $m = 1, 2$ of the top panel of Fig. 1.

n	$P_{\text{on}}^g(0; 0.2)$	$R_{\text{on}}^g(0; 0.2)$
2	5.81	0.467
3	3.54	0.194
4	2.82	0.248
5	2.74	0.213

In Appendix B we present the FORTRAN code used to generate Table I and Fig. 1.

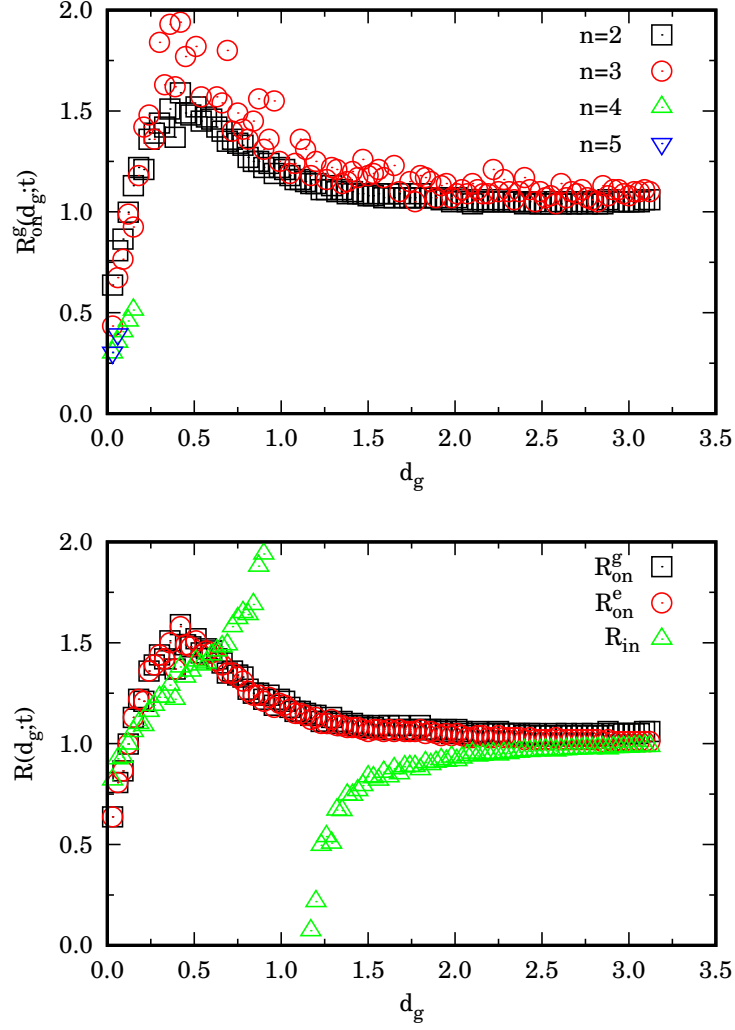


FIG. 1. We show the plot of the ratio $R(d_g; t) = P(m\delta; t)/t\phi(m\delta)$ as a function of $d_g = m\delta$ for $a = 1, \mu_1 = \mu_2 = 1, Z_1 = Z_2 = 1, t = n\tau = 0.2, \delta = 0.03$. On the top panel R_{on}^g for $n = 2, 3, 4, 5$. On the bottom panel the three Coulomb cases $R_{on}^g, R_{on}^e, R_{in}$ for $n = 2$. The angular convolutions in Eq. (2.11) were calculated with a Monte Carlo quadrature keeping the relative error below 40% (this required approximately 4×10^8 MC points in the $n = 5$ case).

III. CUSP CONDITION

On the sphere the cusp condition [1, 4, 5]. can be determined studying the asymptotic behavior as $\mathbf{r}_1 \rightarrow \mathbf{r}_2$ of a vanishing *residual energy* (defined in equation (4.6) of Ref. [1]) for the pair-product approximation to the interaction. On a Sphere as $\mathbf{r}_1 \rightarrow \mathbf{r}_2$ we should compare

$$\left(\frac{\lambda_1}{\sqrt{g(\mathbf{r}_1)}} \nabla_{1\alpha} \sqrt{g(\mathbf{r}_1)} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g(\mathbf{r}_2)}} \nabla_{2\alpha} \sqrt{g(\mathbf{r}_2)} \nabla_2^\alpha \right) P(d_g(\mathbf{r}_1, \mathbf{r}_2); t) = -\phi(\mathbf{r}_1, \mathbf{r}_2). \quad (3.1)$$

As $\mathbf{r}_1 \rightarrow \mathbf{r}_2$, $d_g(\mathbf{r}_1, \mathbf{r}_2) \rightarrow 0$ and we are looking at a local property on the sphere so that we can use the result on the tangent plane at the point $\mathbf{r}_1 = \mathbf{r}_2$ namely

$$\lim_{d_g \rightarrow 0} \frac{\partial P_{on}(d_g; t)}{\partial d_g} = \frac{Z_1 Z_2}{\lambda_1 + \lambda_2}, \quad (3.2)$$

$$\lim_{d_g \rightarrow 0} \frac{\partial P_{in}(d_g; t)}{\partial d_g} = 0, \quad (3.3)$$


```

real*8  th10,ph10,th20,ph20,dg0,de0,ccc
real*8  th1,ph1,th2,ph2,dg,de
real*8  th1p,ph1p,th2p,ph2p,dgp,dep
real*8  rhoid,rho,pot,pcdm,fa,pidm,faid,pcdm2,pidm2
real*8  rhorr,rhoidr
real*8  sigmac,sigmai,avpcdm,avpidm,avpcdm2,avpidm2
real*8  ith1(mni),iph1(mni),ith2(mni),iph2(mni)
integer*8 i,j,k,ni,id,nd,nmc,mmmc
character potk*3,tabf*20

pi=acos(-1.d0)

write(*,*) 'give the timestep tau'
read(*,*) tau
write(*,*) 'mu1 and mu2 in electron mass units'
read(*,*) mu1,mu2
write(*,*) 'Z1 and Z2'
read(*,*) z1,z2
write(*,*) '# timestep iterations, n >= 1, n <= ',mni
read(*,*) ni
ti=(ni+1)*tau
write(*,*) 'imaginary time t = (n+1)*tau =',ti
write(*,*) '# MC steps'
read(*,*) mmmc
write(*,*) 'give the kind of Coulomb potential (in,one,ong)'
read(*,*) potk
write(*,*) '.....'
write(*,*) 'Sphere radius      a =',a
write(*,*) 'time step          tau =',tau
write(*,*) 'g distance step delta =',delta
write(*,*) '.....'
write(tabf,'(I2)') ni
tabf=adjustl(tabf)
tabf='tab-'//trim(potk)//'--'//trim(tabf)//'.dat'
open(unit=8,file=tabf,status='unknown')
write(8,*) '#.....'
write(8,*) '# Sphere radius      a =',a
write(8,*) '# time step          tau =',tau
write(8,*) '# mu1, mu2          =',mu1,mu2
write(8,*) '# z1, z2          =',z1,z2
write(8,*) '# ti              =',ti
write(8,*) '# kind of potential  =',potk
write(8,*) '#.....'

nd = int(pi/delta)

c loop over geodesic distance
do id=1,nd
  th10=id*delta-pi/2
  ph10=0.
  th20=-pi/2
  ph20=0.
  ccc=sin(th10)*sin(th20)+cos(th10)*cos(th20)*cos(ph10-ph20)
  if(ccc.gt.1.) ccc=1.
  if(ccc.lt.-1.) ccc=-1.
  dg0=a*acos(ccc)
  de0=2*a*sin(dg0/2/a)
  pcdm = 0.
  pidm = 0.
  pcdm2 = 0.
  pidm2 = 0.

c MC step
nmc = 0
10 nmc=nmc+1
30 do k=1,ni
  ith1(k)=(rand()-0.5)*pi
  ith2(k)=(rand()-0.5)*pi
  iph1(k)=rand()*2.*pi
  iph2(k)=rand()*2.*pi
enddo
th1=ith1(1)
ph1=iph1(1)
th2=ith2(1)
ph2=iph2(1)
ccc=sin(th1)*sin(th2)+cos(th1)*cos(th2)*cos(ph1-ph2)
if(ccc.gt.1.) ccc=1.
if(ccc.lt.-1.) ccc=-1.
dg=a*acos(ccc)
de=2*a*sin(dg/2/a)
if(dg.lt.epsd)goto 30
rhorr=rho(potk,tau,a,mu1,mu2,z1,z2,
  & th10,ph10,th20,ph20,th1,ph1,th2,ph2,dg,de,dg0,de0)
rhoidr=rhoid(tau,a,mu1,mu2,
  & th1,ph1,th2,ph2,th10,ph10,th20,ph20)
if(rhorr.eq.0..or.rhoidr.eq.0.)goto 30
fa=rhorr
faid=rhoidr
do j=2,ni
  th1=ith1(j-1)
  ph1=iph1(j-1)
  th2=ith2(j-1)
  ph2=iph2(j-1)
  th1p=ith1(j)
  ph1p=iph1(j)
  th2p=ith2(j)
  ph2p=iph2(j)
  ccc=sin(th1)*sin(th2)+cos(th1)*cos(th2)*cos(ph1-ph2)
  if(ccc.gt.1.) ccc=1.
  if(ccc.lt.-1.) ccc=-1.
  dg=a*acos(ccc)
  de=2*a*sin(dg/2/a)
  ccc=sin(th1p)*sin(th2p)+cos(th1p)*cos(th2p)*cos(ph1p-ph2p)
  if(ccc.gt.1.) ccc=1.
  if(ccc.lt.-1.) ccc=-1.
  dgp=a*acos(ccc)
  dep=2*a*sin(dgp/2/a)
  if(dg.lt.epsd)goto 30
  if(dgp.lt.epsd)goto 30
  rhorr=rho(potk,tau,a,mu1,mu2,z1,z2,
    & th1,ph1,th2,ph2,th1p,ph1p,th2p,ph2p,dg,de,dgp,dep)
  & rhoidr=rhoid(tau,a,mu1,mu2,
    & th1,ph1,th2,ph2,th1p,ph1p,th2p,ph2p)
  if(rhorr.eq.0..or.rhoidr.eq.0.)goto 30
  fa=rhorr
  faid=rhoidr
enddo
format(4(d10.3,3x))
close(unit=8)

stop
end

function fact(n)
implicit none
c factorial function
real*8 fact,p
integer i,n

p=1.
do i=1,n
  p=p*dble(i)
enddo
fact=p

return
end

function vr(a,mu,th,ph)
implicit none
c Eq. (2.32) Bastianelli and Corradini EPJC 77, 731 (2017)
real*8 vr,a,mu,th,ph
real*8 pi,m,x,mx,eps
eps=1.d-1
pi=acos(-1.d0)

m =1/a
x =sqrt(th**2.+ph**2.)
mx=m*x
if(x.eq.0.) x=eps
vr=-m**2./6-(5*mx**2.-3*(mx**2.+3)*cos(2*mx))/48/x**2./sin(mx)**2.
if(abs(vr).gt.1/eps) vr=1/eps

return
end

function pot(potk,a,z1,z2,dg,de)
implicit none
c the Coulomb potential
real*8 pot,a,z1,z2
real*8 pi,de,dg
character potk*3
pi=acos(-1.d0)

if(potk.eq.'in') then
  pot=-z1*z2*log(de)
elseif(potk.eq.'one') then
  pot=z1*z2/de
elseif(potk.eq.'ong') then
  pot=z1*z2/dg
else
  write(*,*) 'no pair-potential available !'
  stop
endif

return

```

```

end

function rho(potk,tau,a,mu1,mu2,z1,z2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p,dg,de,dgp,dep)
implicit none
c primitive approximation for rho
real*8 rho,tau,a,mu1,mu2,z1,z2,dg,de,dgp,dep
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 pot,vr,ddx1,ddx2
character potk*3

ddx1=(th1-thip)**2.*(ph1-phip)**2.
ddx2=(th2-th2p)**2.*(ph2-ph2p)**2.

rho = exp(-ddx1/2/mu1/tau-tau*
& (vr(a,mu1,th1,ph1)+vr(a,mu1,thip,phip))/2)
rho =rho*exp(-ddx2/2/mu2/tau-tau*
& (vr(a,mu2,th2,ph2)+vr(a,mu2,th2p,ph2p))/2)
rho =rho*exp(-tau*(pot(potk,a,z1,z2,dg,de)+
& pot(potk,a,z1,z2,dgp,dep))/2)

return
end

function rhoid(tau,a,mu1,mu2,
& th1,ph1,th2,ph2,thip,phip,th2p,ph2p)
implicit none
c primitive approximation for ideal gas rho
real*8 rhoid,tau,a,mu1,mu2
real*8 th1,ph1,th2,ph2,thip,phip,th2p,ph2p
real*8 vr,ddx1,ddx2

ddx1=(th1-thip)**2.*(ph1-phip)**2.
ddx2=(th2-th2p)**2.*(ph2-ph2p)**2.

rhoid = exp(-ddx1/2/mu1/tau-tau*
& (vr(a,mu1,th1,ph1)+vr(a,mu1,thip,phip))/2)
rhoid =rhoid*exp(-ddx2/2/mu2/tau-tau*
& (vr(a,mu2,th2,ph2)+vr(a,mu2,th2p,ph2p))/2)

return
end

```

AUTHOR DECLARATIONS

Conflicts of interest

None declared.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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